SARTHAK DRAVID

+91 7674918895 Sarthak.dravid2021@gmail.com Sarthak Dravid Hyderabad, Telangana

EDUCATION

Manipal Institute of Technology(MAHE) Jul. 2023- Jul 2027

B.Tech Cyber Physical Systems Engineering

- **Current GPA: 8.19**
- **CGPA: 7.55/10**
- **Relevant coursework:** Data structures and Algorithms; Computer Architecture and Organization; Sensor Technology; Communication systems; Digital transmission; Systems Design

TECHNICAL SKILLS

Design, Modeling, and Simulation:

- **3D modeling and reconstruction:** Blender, AutoCAD, freeCAD, Solidworks, Meshroom, Metashape, pix4dmapper, polycam
- **Graphics and Animation:** Photoshop, Krita, Canva, FireAlpaca , Inkscape, Davinchi Resolve
- **Simulation:** MATLAB & Simulink
- **Video editing**

Programming and electronics:

- **Computer Languages:** Python, Game development in C++,LaTeX (with prev. experience in Java, C, and HTML)
- **Game engines:** Unity, GameMaker (with some experience with godot and unreal engine)
- **Microcontrollers:** Raspberry Pi, Arduino, 8051 (with some experience with ARM and ESP 32)

AI and Machine Learning:

- **Libraries**: NumPy, Pandas, Scikit-Learn, OpenCV
- **Deep Learning Frameworks:** Tensorflow
- **Transformers and NLP:** GPT, Hugging Face, Llama

PROFESSIONAL EXPERIENCE

Nakshatra Mapping Solutions 2025 - PRESENT

Co-Founder, Head of innovation

- Co-founded Nakshatra and led R&D and technical development, overseeing the development of 5+ geospatial products
- Deployed a live interactive web map for Revels'25 in collaboration with the Student Council
- Worked with advanced 3D reconstruction techniques specialising in real world application
- **Second Runner-up & "Best Pitch" Award, Ground Reality '25 at BITS Pilani Hyderabad out of 200+ teams**

Open Horizon Robotics 2024- PRESENT

3D engineer

- Created 10+ high-resolution 3D building models using Meshroom and Blender
- Performed peer code reviews and merged >50 pull requests with zero critical defects.
- Assisted in developing an open-source autonomous drone
- Designed custom merchandise for the Manipal chapter

CAD designer 2025

Freelance

- Collaborated with clients to develop precise and professional-grade technical CAD drawings tailored to their unique design specifications

Mars Rover Manipal 2023 - 2024

Junior taskphase member- Research AI

- Worked on multiple tasks demonstrating AI, data analysis, ML from scratch with only numpy and pandas in Jupyter lab and prepared IEEE compliant research reports for every task

CERTIFICATIONS

- Python for Data Analysis: Pandas & NumPy
- Computer Vision Basics by University at Buffalo
- Wireless Communications Onramp by MATLAB
- Signal Processing Onramp by MATLAB

PROJECTS

Slides Extactor

A simple tool to obtain study material easily

- Built a tool to extract slide content from lecture videos using OpenCV and TensorFlow, achieving over 90% slide detection accuracy
- Packaged tool as a CLI utility and guided fellow peers through installation

Custom Handheld Metal detector

Sensor Technology mini project

- Built a 25x7x4 cm portable Arduino-based metal detector tested across 15+ metallic samples with 75% accuracy
- Differentiated between ferromagnetic and diamagnetic metals

Report Card Management system

DSA mini project

- Developed a text-based system with CRUD functionality
- Implemented multi-key sorting/searching and automatic CGPA computation

Line Following Robot

MC mini project

- Designed and programmed a simple line-following robot using the 8051 microcontroller
- Integrated two IR sensors for real-time path detection and correction based on surface reflectivity.
- Implemented logic in assembly to control motor directions and maintain alignment along a predetermined path.

Navi(in progress)

AR based indoor navigation system

- streamlining AR navigation pipeline for effective and efficient implementation in practical situations

Twinify(in progress)

Optimized 3D Reconstruction Model for GIS Integration

- Prototyping a lightweight ML-driven 3D reconstruction model for GIS data pipelines, aiming to reduce processing time by 30% vs. traditional photogrammetry